Mapping O*NET Tasks to ISCO Occupations using Text Similarity

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Abstract

O*NET task data are widely used to measure technology exposure and skill demand, but O*NET is based on the U.S. Standard Occupational Classification (SOC) while international labour statistics use ISCO. Existing SOC–ISCO concordances operate at the occupation level, ignoring within-occupation task variation. This paper constructs a crosswalk from O*NET tasks to 4-digit ISCO-08 occupations using text similarity and ESCO information, covering 50 O*NET releases from v15.1 to v30.3 under the SOC 2010 and SOC 2018 taxonomies. Across SOC 2018 releases, 18,796–19,281 tasks are assigned to 401–402 of 436 ISCO-08 groups, with 81.6–81.7% exact agreement and 94.1–94.2% major-group agreement against institutional crosswalks. The crosswalk enables task-based measures of automation, skills, and AI exposure to be applied to internationally comparable labour statistics.

Keywords: Occupational classification; O*NET; ISCO-08; Task-level crosswalk; Text similarity; Labour market

JEL codes: J24; J44; C88; O33; C80

1 Introduction

Many studies in labour economics use O*NET task data to measure technology exposure, skill demand, and occupational change (Autor et al., 2003; Acemoglu and Autor, 2011; Frey and Osborne, 2017). More recent datasets measuring exposure to artificial intelligence are also published at the O*NET task level (Eloundou et al., 2024), based on both the SOC 2010 and SOC 2018 versions of O*NET.

O*NET is organised under the U.S. Standard Occupational Classification (SOC), while most international employment statistics are indexed using the International Standard Classification of Occupations (ISCO) (International Labour Office, 2012). Existing SOC–ISCO concordances operate at the occupation level (O*NET Center, 2022; ESCO Secretariat, 2022; Bureau of Labor Statistics, 2015; Matysiak et al., 2024). Two of these were constructed using text similarity (O*NET Center, 2022; Matysiak et al., 2024), but all four match occupation titles rather than individual tasks, averaging task content within each SOC code and discarding within-occupation variation that is central to task-level exposure measures.

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This paper builds a direct task-level crosswalk from O*NET to 4-digit ISCO-08 occupations using text similarity. The method compares the text of O*NET task statements augmented with the SOC occupation title and Detailed Work Activities (DWAs) on one side with occupational descriptions from ISCO-08 and ESCO on the other side, allowing each task to be matched to the most similar ISCO-08 occupation group. Crosswalk versions are provided for all O*NET releases from v4.0 to v30.3, spanning SOC 2018, SOC 2010, and earlier classifications.

The crosswalk covers between 401 and 404 of the 436 ISCO-08 unit groups across all 50 SOC 2010 and SOC 2018 releases (v15.1–v30.3), with task counts ranging from 18,783 to 19,735 and mean cosine similarity stable at 0.715–0.718. The mapping is also highly stable across O*NET releases. Among 16,049 tasks shared between O*NET 29.2 (SOC 2018) and O*NET 25.0 (SOC 2010), assignments agree in 97.7% of cases. The crosswalk allows existing task-based measures of automation, skills, and AI exposure to be applied directly to internationally comparable labour statistics.

2 Data

2.1 Occupational data

The task data source is O*NET, a database of occupational information maintained by the U.S. Department of Labor and updated quarterly. Crosswalk files are provided for all 63 releases from v4.0 to v30.3. Validation uses O*NET 29.2 (the most recent SOC 2018 release at the time of parameter selection), which contains 18,796 task statements. Each statement describes a discrete work activity within a specific SOC occupation. Each task is further linked to one or more Detailed Work Activities (DWAs), a cross-occupation vocabulary of 2,082 granular activity descriptors used as an additional embedding signal.

The target classification is ISCO-08 at the four-digit unit-group level (International Labour Office, 2012). ISCO-08 is the international standard used in most cross-country labour statistics. Official ISCO-08 task descriptions are available for 435 of the 436 unit groups; the remaining group (ISCO 1113, Traditional Chiefs) has no U.S. counterpart.

Data from ESCO are also used (European Commission, 2019). ESCO includes approximately 3,000 occupation concepts linked to ISCO-08 unit groups and adds skill and competence information that complements the official ISCO-08 task descriptions.

2.2 Institutional crosswalks for validation

Four existing SOC–ISCO-08 concordances are used for validation only. For SOC 2018: the O*NET–ESCO crosswalk (O*NET Center, 2022) and the ESCO–SOC 2018 crosswalk (ESCO Secretariat, 2022). For SOC 2010: the BLS SOC 2010–ISCO-08 crosswalk (Bureau of Labor Statistics, 2015) and the ESCO–O*NET crosswalk (Matysiak et al., 2024).

3 Method

3.1 Overview

The pipeline maps each O*NET task to a single ISCO-08 unit group via cosine similarity (Figure 1). All texts are encoded with `all-mpnet-base-v2` sentence embeddings (Reimers and Gurevych, 2019).

3.2 Task and occupation representations

Each O*NET task is represented using the task statement text, the title of the SOC occupation to which the task belongs, and optionally the associated Detailed Work Activity (DWA) labels. All text components are encoded by the `all-mpnet-base-v2` sentence encoder into unit-normalised real-valued vectors: each embedding $e \in \mathbb{R}^{768}$, $\|e\|_2 = 1$. All mixing weights are scalars $w \in [0, 1]$, so every composite representation is a convex combination of unit vectors and is itself a vector in $\mathbb{R}^{768}$. Task embeddings are combined as

$$q = (1 - w_{\text{soc}})\big[w_{\text{dwa}}\, e_{\text{dwa}} + (1 - w_{\text{dwa}})\, e_{\text{task}}\big] + w_{\text{soc}}\, e_{\text{soc}}, \tag{1}$$

where e_{task} is the embedding of the O*NET task text, e_{dwa} is the embedding of the associated DWA labels, and e_{soc} is the embedding of the SOC occupation title. Including occupation context helps distinguish tasks that use similar language in different occupational settings.

Each ISCO-08 occupation is represented using both official ISCO-08 task descriptions and ESCO occupation information. The ISCO component combines ISCO task descriptions with the ISCO occupation title,

$$e_{\text{ISCO}} = w_{\text{isco,task}}\, e_{\text{isco,task}} + (1 - w_{\text{isco,task}})\, e_{\text{isco,title}}. \tag{2}$$

The ESCO component combines ESCO occupation descriptions with ESCO skill texts,

$$e_{\text{ESCO}} = w_{\text{occ}}\, e_{\text{esco,occ}} + (1 - w_{\text{occ}})\, e_{\text{esco,skill}}. \tag{3}$$

Where multiple ESCO occupations map to the same ISCO-08 unit group, their embeddings are averaged.

The final occupation representation combines both components,

$$e_{\text{occ}} = w_{\text{isco}}\, e_{\text{ISCO}} + (1 - w_{\text{isco}})\, e_{\text{ESCO}}. \tag{4}$$

3.3 Assignment rule

For each task query q, cosine similarity $\text{sim}(q, e_{\text{occ}}) = q \cdot e_{\text{occ}} \in [-1, 1]$ (dot product of unit-normalised vectors) is computed against all ISCO-08 occupation representations. Each task is assigned to the occupation with the highest similarity score. A minimum similarity threshold and a margin between the best and second-best match are used to assess assignment confidence;

all tasks receive an assignment regardless of confidence level (embedding ≈19,000 tasks takes ≈30 minutes on a standard CPU, ≈2 GB peak; embeddings are cached by hash so subsequent releases complete in minutes and FAISS retrieval over 436 targets adds negligible overhead).

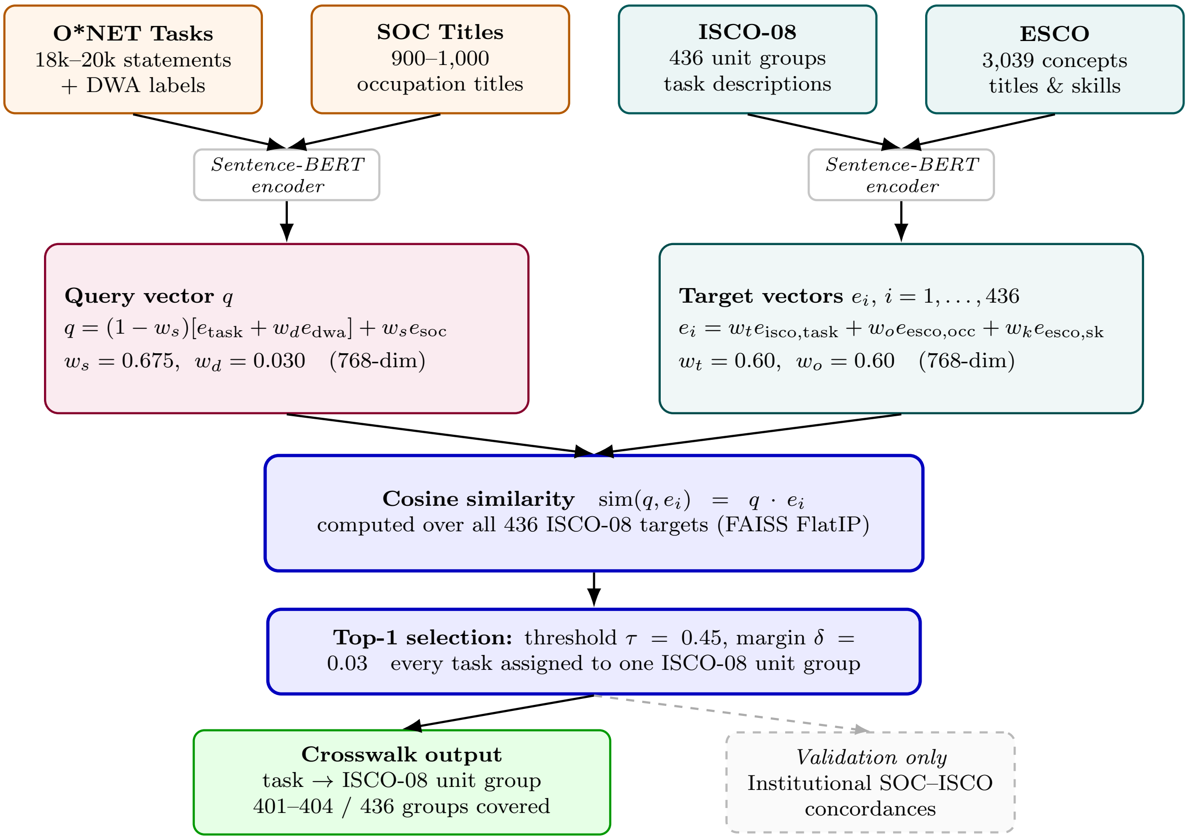

Figure 1: Task-level mapping from O*NET to ISCO-08.

Notes: Task representations combine O*NET task text, DWA labels, and SOC occupation title. Occupation representations combine official ISCO-08 task descriptions with ESCO occupation titles and skills. Similarity matching assigns each task to one ISCO-08 unit group. Institutional SOC–ISCO-08 crosswalks are used only for validation.

4 Validation

Because no gold standard exists for direct task-to-ISCO mapping, validation uses institutional occupation-level SOC–ISCO crosswalks. Task-level assignments are aggregated to the occupation level and compared with these concordances as an external benchmark.

Agreement with institutional crosswalks is high. Across all 23 SOC 2018 releases (O*NET 25.1–30.3), 81.6–81.7% of tasks are assigned to the same ISCO-08 unit group as in at least one institutional crosswalk, while 94.1–94.2% agree at the major-group level. Across all 27 SOC 2010 releases (O*NET 15.1–25.0), exact agreement is 63.2% and major-group agreement 82.8%, constant across the full era. The lower SOC 2010 figures reflect the weaker institutional crosswalks available for that taxonomy: fewer occupations are covered, and one of the two reference crosswalks is itself derived using semantic similarity, reducing its independence as a benchmark.

Among the 16,049 tasks shared between O*NET 29.2 (SOC 2018) and O*NET 25.0 (SOC 2010), assignments agree in 97.7% of cases.

Agreement rates are robust to the choice of reference crosswalk; detailed tables are in Online Appendix C. 95% Wilson confidence intervals for all reported SOC 2018 agreement rates are within ±0.6 percentage points ($n \approx$ 18,800); SOC 2010 rates are within ±0.7 pp ($n \approx$ 19,700).

Results are stable across parameter choices and releases (Online Appendix A, C). Two alternative encoders (`bge-large-en-v1.5`, `gte-large`) produce exact agreement rates of 83.7%–84.7%, within 3.0 pp of the main specification (Online Appendix F). Examples and mismatches are in Online Appendix E.

A key limitation of this benchmark is that it is indirect: it confirms occupation-level consistency with institutional concordances but cannot evaluate within-occupation task-level precision, for which no ground truth exists.

5 Parameter Selection

The main specification uses $w_{\text{dwa}} = 0.030$, $w_{\text{soc}} = 0.675$, $w_{\text{isco}} = 0.80$, $w_{\text{isco,task}} = 0.60$, and $w_{\text{occ}} = 0.60$. The similarity threshold is set to 0.45 with margin parameter 0.03. Each task is assigned to the single most similar ISCO-08 unit group.

Parameter values are chosen using unsupervised criteria reflecting the objectives of the mapping. Candidates are scored on ISCO-08 coverage, mean similarity, and task concentration across groups; excessive concentration is penalised as a sign of poor semantic differentiation. The selected configuration's low concentration share (2.9%) reflects structural SOC–ISCO-08 differences: the two groups above the threshold each draw over 75% of their tasks from a single SOC major group (Online Appendix E).

Three qualitative patterns emerge. First, the selected DWA weight is near zero ($w_{\text{dwa}} = 0.030$); the ablation study confirms that removing DWA entirely has no measurable effect on validation agreement, while large weights reduce it, indicating that DWA text adds little beyond the task statement at this scale. Second, a high SOC occupation title weight ($w_{\text{soc}} = 0.675$) produces more stable assignments, reflecting that occupational context is a strong signal for cross-classification matching. Third, combining ISCO-08 task descriptions with ESCO occupation text improves retrieval relative to using either source separately, indicating that the two sources provide complementary information.

Sensitivity analysis in Online Appendix A confirms stability across reasonable variation in parameter values. Online Appendix G presents an ablation study: removing the SOC occupation title reduces exact agreement by 38 percentage points (81.7% to 43.6%), while removing ESCO reduces it by 2.2 pp; DWA labels have no measurable effect at $w_{\text{dwa}} = 0.030$.

6 Conclusion

The crosswalk maps O*NET task statements to ISCO-08 four-digit unit groups across 50 releases from v15.1 to v30.3. Across SOC 2018 releases, 18,796–19,281 tasks are assigned to 401–402 groups with 81.6–81.7% agreement; across SOC 2010 releases, 18,783–19,735 tasks to 403–404 groups (variation reflects additions and removals of SOC occupations between releases). The crosswalk enables task-based measures of automation, skill content, and AI exposure to be applied to internationally comparable labour statistics without occupation-level aggregation. Researchers should prefer task-level matching when within-occupation variation matters; an occupation-level concordance suffices when all tasks in a SOC occupation receive the same value. One structural limitation applies: ISCO groups without a U.S. counterpart cannot be matched (e.g., ISCO 1113). Replication materials are provided in the Online Appendix.

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Data Availability

The crosswalk dataset and replication materials are available at `https://doi.org/10.5281/zenodo.21932817`.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used Claude (Anthropic), ChatGPT (OpenAI), Claude Code (Anthropic), and Codex (OpenAI) for language editing and writing assistance, and for code generation in the development of the text similarity pipeline. After using these tools, the author reviewed and edited all content as needed and takes full responsibility for the content of the published article.

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